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卷册检索号 SEARCH NO. OF THE VOLUME
MLHVDC-ENG-CET-DD-GL-0104 30-B003701S-Z0104

第 1 页 PAGE 共 1 页 OF

图纸文件目录(首页)

CATALOG OF DRAWING DOCUMENTS (HEAD PAGE)

拉合尔±660kV换流站 工程 PROJECT ±660kV Lahore Converter Station

施工图 阶段 DESIGN STAGE Detail Design

总图 专业 SPECIALITY General Layout 第 01 卷 VOLUME 第 04 册 BOOK

卷册 名称 Volume name 围栏及边坡图 Fence and Slope Chart

批 准 APPROVED BY	
审 核 REVIEWED BY	
设计/勘测 DESIGNED BY/ SURVEYED BY	
图纸 4 张 DRAWING NOS.	打印件 0 本 PRINTED COPIES NOS.

序号 No.	图 号 DRAWING NO.	文件名称 DOCUMENT TITLE	版本 EDITION	张 NOS.	本 NOS.	备注 REMARKS
1	30-B003701S-Z0104-01	全站围栏、边坡平面布置图 Fence and Slope protection Layout	A	1		
2	30-B003701S-Z0104-02	护坡详图 Details of Slope Protection	A	1		
3	30-B003701S-Z0104-03	大门详图 Details of Gate	A	1		
4	30-B003701S-Z0104-04	围栏详图 Details of Fence	A	1		

注:

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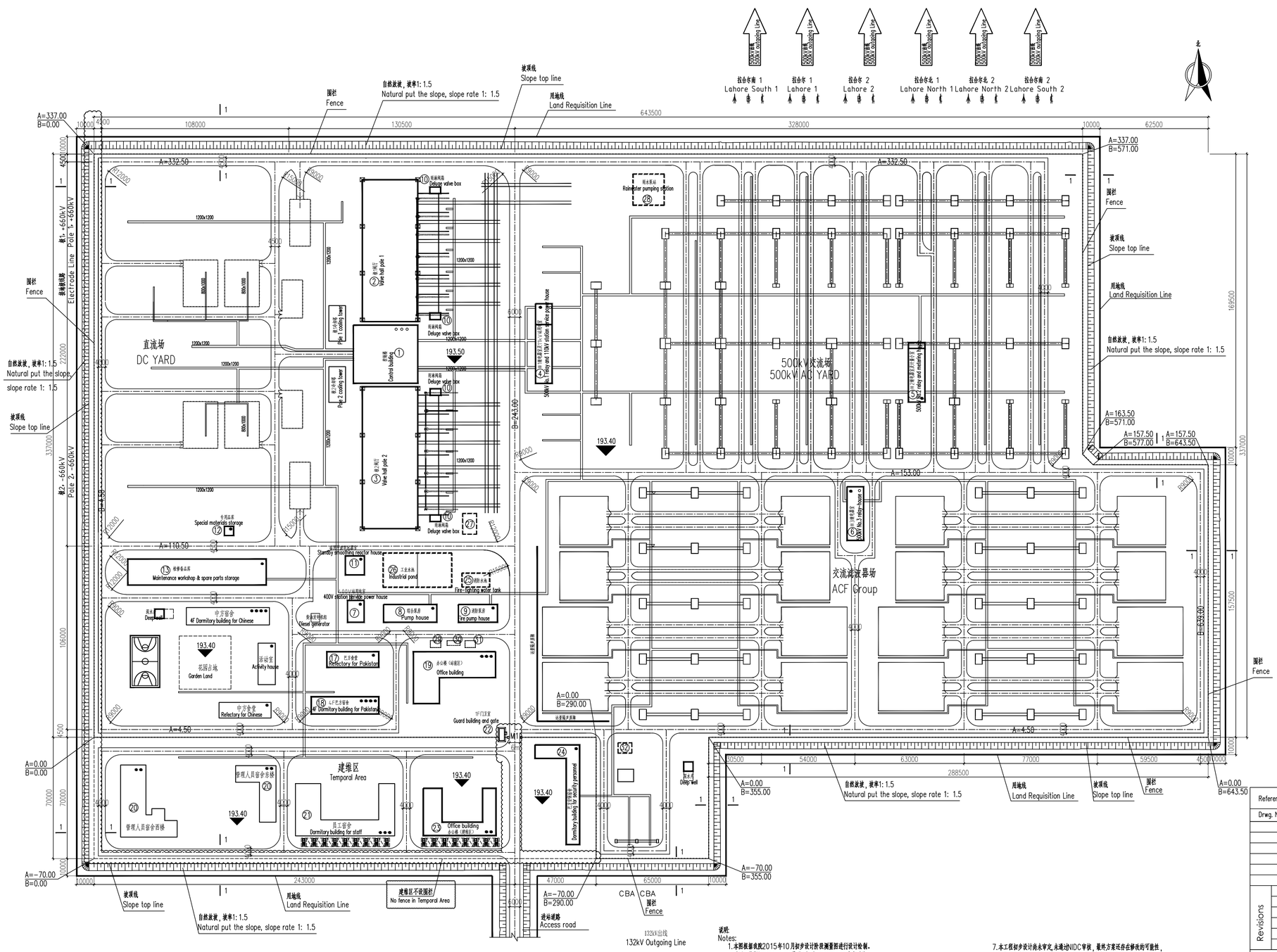
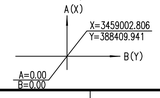


图例:
Legend:

建筑物 Building
道路 Road
围栏 Fence
地下建筑 Building underground
架构 Substaion frame
电缆沟 Cable trench
护坡 Slope protection

说明:
Notes:

1.本图根据前说既2015年10月初步设计阶段测量图进行设计绘制。
采用巴基斯坦UTM-43N平面坐标系，EGM2008高程系统。
1.This drawing is designed based on preliminary design stage topographic map provided by ECEPDI in October 2015.
Coordinate system: UTM 43N, Elevation system:EGM2008.
2.相对平面坐标以米(m)为单位，尺寸以毫米(mm)为单位。
2.Relative coordinates system is measured in meter.
Dimensions are measured in millimeters.
3.站地平均设计标高为193.40m。
3.Site level is equal to 193.40m.
4.由于站地内部平标高(192.4m)低于站地外边坡坡顶标高(193.4m)，施工时请注意次界处处理问题。
4. Because the Initial elevation of site (192.4) is lower than the slope height of the side slope (193.4), please pay attention to the junction treatment.
5.站区西侧及建维区围栏位置应合理布置，施工前请联系设计单位。
5.The position of fence in west side and temporal area may need to be modified, please contact the design unit before construction.
6.本图建筑(A、B)坐标与UTM-43N平面坐标系关系如下所示:
6.The relationship between relative coordinate system and UTM-43N system is as below:



7.本工程初步设计尚未审定，未通过NIDC审核，最终方案还存在修改的可能性，施工前请联系业主与设计确认，由此引起的方案变更设计不予负责。
7.Because of preliminary design documents are not approved by NIDC, the project solution has prossibility to be modified.
The design contractor has no responsibility for the project solution changed, if the construction contractor not get authorized by project owner and design contractor before construction started.
8.根据初论原则审查会沟通结论，本站西侧按围墙外新增5m进行征地平，围墙位置暂保持不变，最终方案待初设审定后修改。
8.According to the conclusion of principle examination meeting for preliminary design, site level work is formed as 5m away from west side of the fence, the position of fence temporarily stay unchanged, the final scheme is pending for final approval of preliminary design.
9.本图以CP-3、CP-10、CP-6、CP-8为定位点，坐标分别为:
9.The drawing uses point CP-3, CP-10, CP-6 and CP-8 as locating points, and the corresponding coordinates are as below:

CP-3	X(m)=3458638.678	CP-10	X(m)=3458726.757
191.166	Y(m)=388554.185	191.521	Y(m)=389586.869
CP-6	X(m)=3459413.990	CP-8	X(m)=3459420.705
191.354	Y(m)=389040.061	191.066	Y(m)=388398.196

10.1-1、2-2截面详见B003701S-Z0104-02图。
10.1-1 cross section and 2-2 cross section are shown in Figure B003701S-Z0104-02.

全站围栏、边坡平面布置图
Fence and Slope protection Layout

Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦默蒂亚里-拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 State Grid 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

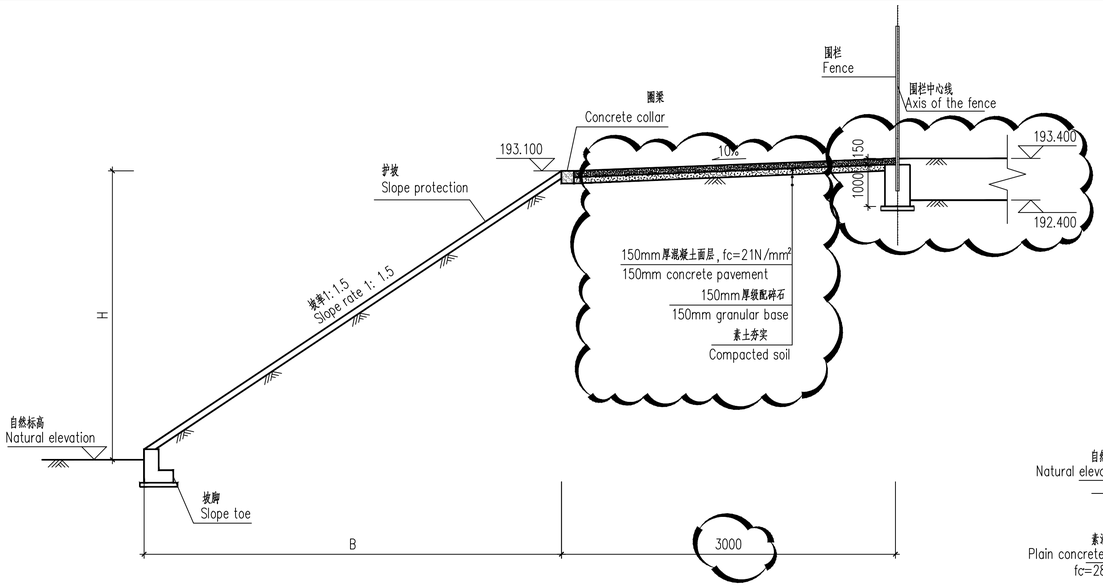
M L H V D C - E N G - C E T - D D - G L - 0 1 0 4 - 0 0 1 A

	姓名 Name	日期 Date
设计 Designed	孟彩	
校核 Checked	刘俊	
审核 Reviewed	吕吉晴	
批准 Approved	李柏青	

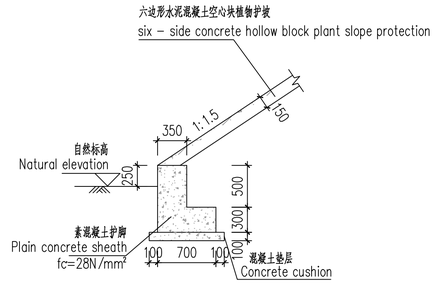
全站围栏、边坡平面布置图
Fence and Slope protection Layout

比例 Scale: 1:1200
图号 DWG NO.: 30-B003701S-Z0104-01
版本 REV.: A

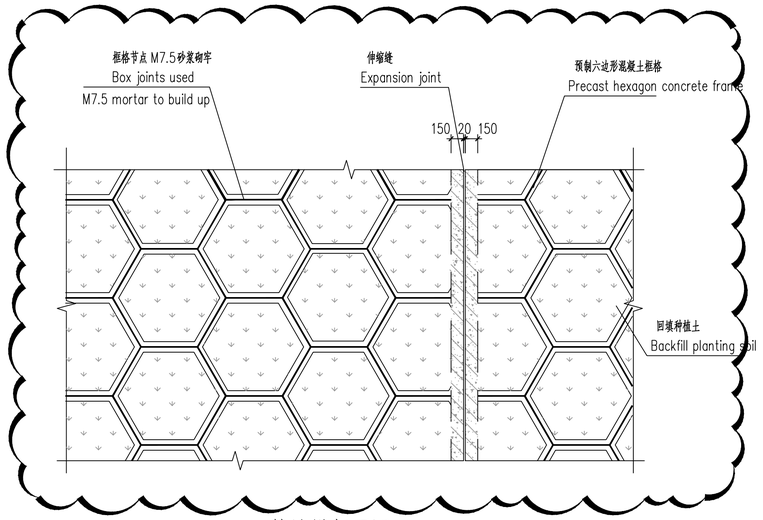
A1(841x594)



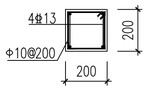
1-1 1:100



坡脚详图
detail of Slope Toe 1:50



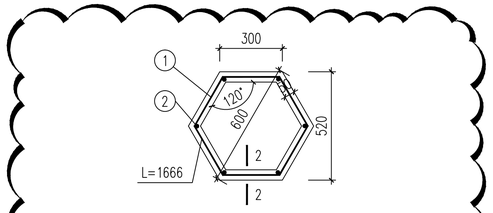
护坡横断面图
Cross section of slope protection 1:50



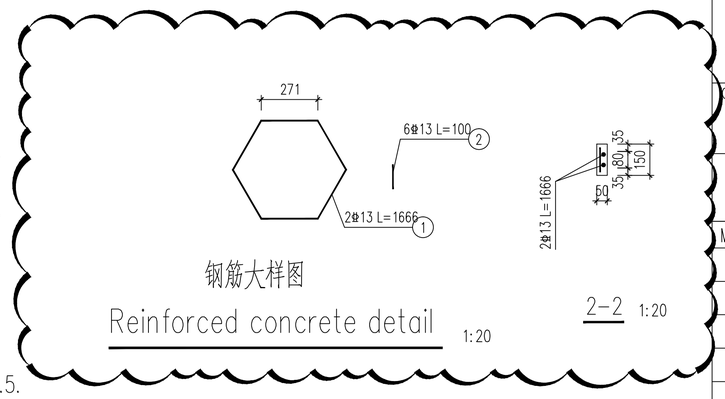
A-A 1:20
圈梁

说明:
Notes:

1.相对平面坐标以米(m)为单位，尺寸以毫米(mm)为单位。
1.Relative coordinates system is measured in meter. Dimensions are measured in millimeters.
2.场地平均设计标高为193.40m。
2.Site level is equal to 193.40m.
3.垫层的混凝土强度要求不低于fc'=14MPa；其他部分混凝土强度要求不低于fc'=21MPa。
3.The concrete strength of cushion layer is not below 14Mpa, and the rest of the concrete strength of cushion layer is not below 21MPa.
Φ: Grade 40，最小抗拉强度≥420MPa；Φ: Grade 60，最小抗拉强度≥620MPa。
Φ: Grade 40, the minimum yield strength is not below 240MPa,
Φ: Grade 60, the minimum yield strength is not below 620MPa.
4.护坡采用六边形水泥混凝土空心块植物护坡，厚度150mm，坡顶设置圈梁，坡脚设置素混凝土坡脚。
4. Slope protection using six - side concrete hollow block plant slope protection, with a thickness of 150 mm, the longitudinal reinforcement of the gridiron skeleton is anchoed in the concrete collar of the slope top and the toe of the concrete.
5.坡面需绿化，表层土种草籽,草种应根据气候区划进行选型，要求有优良的抗逆性，并采用两种以上的草种进行混播。
5.Slope surface should be green, surface soil is used for grass seed, grass should be selected according to the climate division, requires excellent resistance, and requires more than two species of grass to be mixed.
6.六边形水泥混凝土空心块植物护坡与坡脚每隔15m设置伸缩缝一道，宽20mm，沥青麻丝满缝隙填塞。
6. Six - side concrete hollow block plant slope protection and foot slope at the foot of each 15m set a joint expansion joint,with 20 mm wide and full gap caulking of the asphalt.
7.边坡基层做法要求回填土的压实系数≥0.95。
7.The slope base practice requires compaction coefficient of backfill soil greater than 0.95.
8.伸缩缝两侧设置格栅梁，尺寸及配筋详见A-A剖面。
8. The grille beam is arranged on both sides of the expansion joint, the dimensions and reinforcement are shown in A-A section.
9.材料采用水泥混凝土预制块，混凝土强度不低于fc'=21MPa，砂浆强度不低于M7.5。
9. The material shall adopt cement concrete precast block, the strength of concrete shall not be lower than fc'=21MPa, and the strength of mortar shall not be lower than M7.5.



六边形框格平面大样图
Hexagonal box - frame graphic detail 1:20



钢筋大样图
Reinforced concrete detail 1:20

2-2 1:20

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦戴蒂亚里-拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: Detail Design

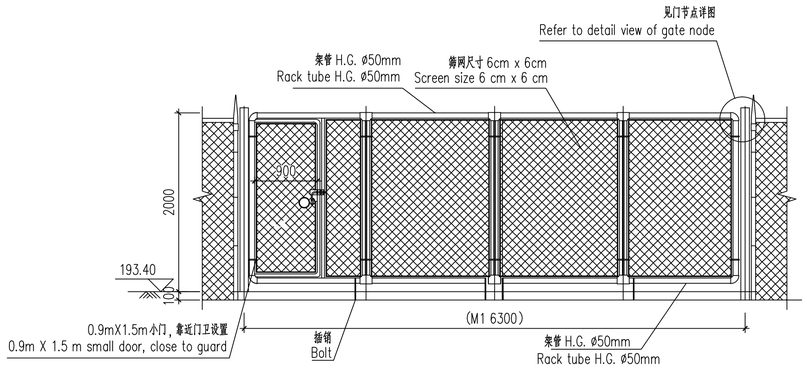
M L H V D C - E N G - C E T - D D - G L - 0 1 0 4 - 0 0 2 A

	姓名 Name	日期 Date
设计 Designed		
校核 Checked		
审核 Reviewed		
批准 Approved		

护坡详图
Details of Slope Protection

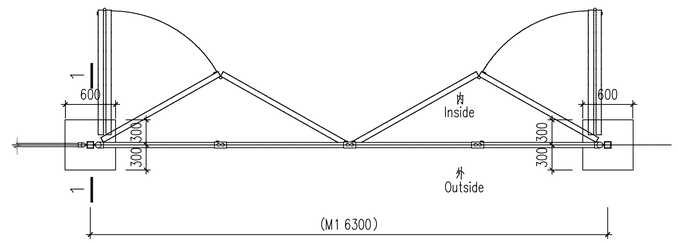
比例 Scale | 图号 DWG NO. 30-B003701S-Z0104-02 | 版本 REV. A

A2(594x420)



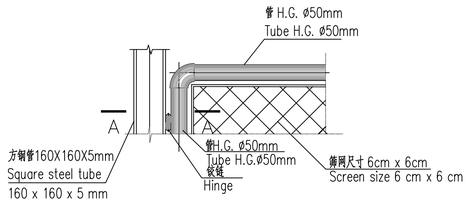
M1大门立面图

Gate M1 elevation chart 1:50



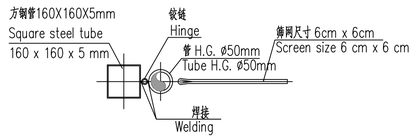
M1大门平面图

M1 Gate plane graph 1:50

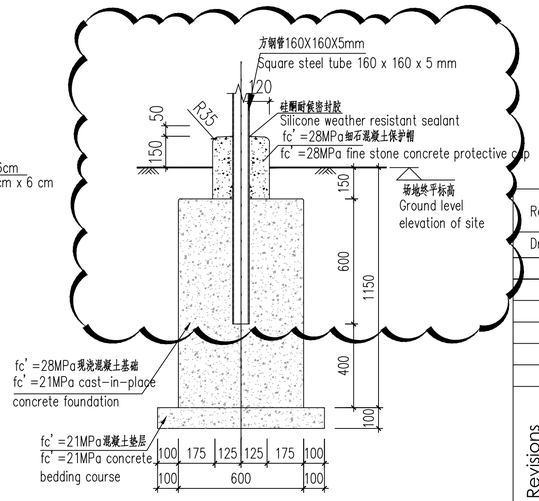


门节点详图

Detail view of gate node 1:20



A–A 1:20



1–1 1:20

说明:

Notes:

1.相对平面坐标以米(m)为单位，尺寸以毫米(mm)为单位。

1.Relative coordinates system is measured in meter.
Dimensions are measured in millimeters.

2.场地平均设计标高为193.40m。

2.Site level is equal to 193.40m.

3.格栅围栏采用钢丝网围栏，由业主监理确认后施工。

3. The grille fence shall adopt the steel wire mesh fence, and shall be constructed after confirmation by the owner's supervision.

4.所有的钢构件除标明外均采用Q235钢；焊条应是E43XX,而且焊缝高度除注明外是5mm。

4.All steel materials shall be grade Q235(yield strength is 235MPa) except otherwise noted.Electrode shall be E43XX and height of welding seam is 5mm except otherwise noted.

Reference Drawings

Drwg. No.	Title

Revisions	Name	Date	Notes

巴基斯坦默蒂亚里–拉合尔±660kV直流输电工程

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拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - E N G - C E T - D D - G L - 0 1 0 4 - 0 0 3 A

	姓名 Name	日期 Date
设计 Designed		
校核 Checked		
审核 Reviewed		
批准 Approved		
比例 Scale		图号 DWG NO.

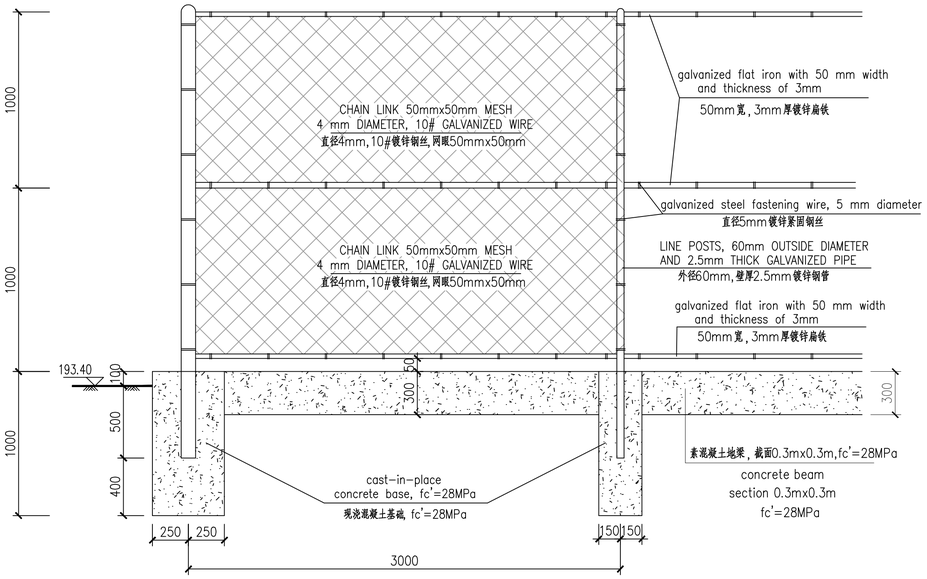
大门详图

Details of Gate

30-B003701S-Z0104-03

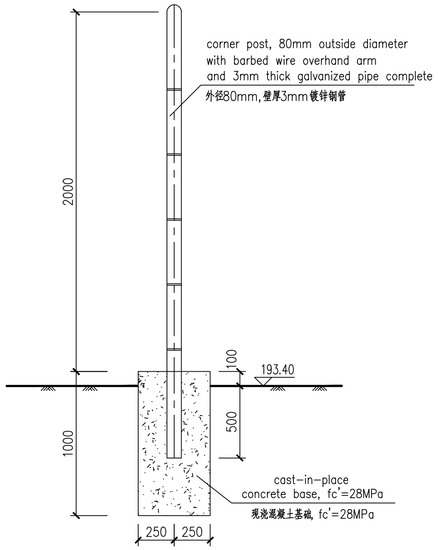
版本 REV. A

A2(594x420)



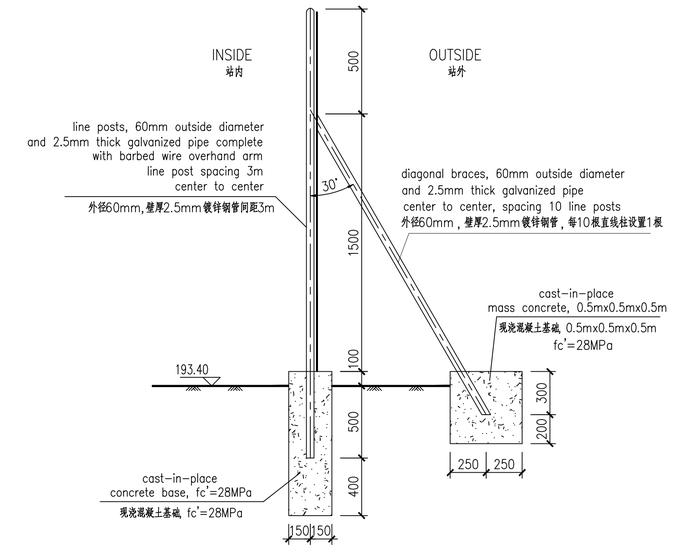
SECURITY FENCE ELEVATION

安全围栏立面图



TYPICAL CORNER POST DETAIL

转角柱详图



TYPICAL LINE POST DETAIL

直线柱详图

说明:

1. 图中除标注外,所有尺寸单位为mm。
2. 焊条采用E43XX,焊缝高度5mm。
3. 除注明外,所有钢材采用Q235(屈服强度235MPa)。
4. 混凝土抗压强度应不低于28MPa.

NOTES:

1. All dimensions are in millimeter, except otherwise noted.
2. Electrode shall be E43xx and height of welding seam is 5mm except otherwise noted.
3. All steel materials shall be grade Q235(yield strength is 235MPa) except otherwise noted.
4. The concrete compressive strength shall not less than 28MPa.

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

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拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - E N G - E C I - D D - G L - 0 1 0 4 - 0 0 4 A

	姓名 Name	日期 Date
设计 Designed	孟红	
校核 Checked	刘晓	
审核 Reviewed	吴建强	
批准 Approved	杨之芳	

围栏详图

Details of Fence

比例 Scale		图号 DWG NO.	B003701S-Z0104-04	版本 REV.	A

A1(841x594)